parallel to our brane and so spread into just two transverse dimensions, whence gravity obeys the observed inverse square law.[39]

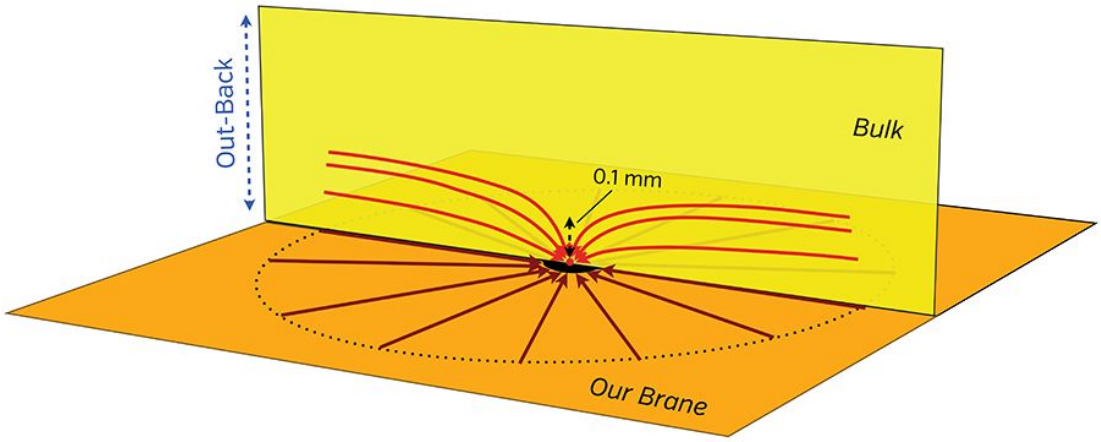


Fig. 23.6. If the bulk experiences AdS warping, then gravitational force lines bend parallel to our brane, because far from the brane there is very little volume in which to spread. *[Patterned on a figure in Lisa Randall's* Warped Passages *(Randall 2006).]*

The AdS Sandwich: Plenty of Room in the Bulk

Sadly, the precipitous shrinkage of distances parallel to our brane, as you move outward, makes the bulk's volume above and below our brane too small for Cooper and his tesseract, and too small for any other human activity in the bulk. I recognized this problem way back in 2006, when *Interstellar* was in its infancy, and I quickly conceived a solution for my science interpretation of the movie: Confine the AdS warping to a thin layer around our brane, a "sandwich." Do so by placing two other branes, confining branes, alongside ours (Figure 23.7). In the sandwich between these branes, the bulk suffers AdS warping. Outside the sandwich, the bulk is totally unwarped. So there is all the volume any sci-fi writer could want, outside the sandwich, for bulk-based adventures.